RAMESH KALIAPPAN

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PROFILE SUMMARY:

Experienced Java Developer with expertise in Spring Boot, Spring Cloud Data Flow and Microservices Architecture, specializing in ONF TAPI-based implementations. Strong proficiency in Kafka-based event-driven systems, ensuring seamless communication between distributed services. Skilled in deploying containerized applications using Docker and Kubernetes, enabling scalable and resilient cloud-native solutions. Proven experience in leading a team of engineers, mentoring junior developers, assigning tasks and ensuring timely project delivery. Expertise in System Design, Scalability, Distributed Systems and Performance Optimization.

EDUCATION::

College of Engineering Guindy, Anna University Chennai, India
Bachelor of Technology in Information Technology May 2014
CGPA: 7.85/10.0

TECHNICAL SKILLS:

Programming: Java, Javascript
Messaging & Streaming: Apache Kafka
Containerization & Orchestration: Docker, Kubernetes
Frameworks: Hibernate, Kafka, Microservices, Spring Boot
Tools: Eclipse, IntelliJ IDEA, Redmine, GitHub
Microservices & Cloud: Spring Boot, Spring Cloud Data Flow, REST APIs, ONF TAPI
Operating Systems: Linux, Windows
Databases: MySQL, Oracle, OrientDB
Version Control: Git, SVN
DevOps & CI/CD: Jenkins, GitLab

PROFESSIONAL EXPERIENCE:

NMSWorks Software Private Limited Chennai, India
Lead Engineer February 2022 – Present

- Designed and implemented ONF TAPI-based microservices using Spring Boot and Spring Cloud Data Flow, enabling SDN-based transport network orchestration.
- Developed an event-driven architecture using Kafka, ensuring efficient asynchronous communication across distributed services.
- Containerized and deployed microservices using Docker and Kubernetes, achieving scalability, resilience and high availability.
- Led a team of 4 engineers, driving development activities, conducting code reviews, mentoring juniors and coordinating task execution.
- Collaborated with cross-functional teams to ensure on-time and quality delivery of enterprise telecom modules.

NMSWorks Software Private Limited Chennai, India
Senior Software Engineer May 2017– February 2022

- Led the Resource Allocation Framework development for PDH, SDH, Ethernet, WDM and OTN technologies across projects like Google, Airtel, SpaceWorld and PGCIL.
- Spearheaded network discovery and real-time reconciliation, improving inventory accuracy and network visibility.
- Conducted technical training sessions on Optical Transport Network Management Systems (OTNMS) for new engineers.

NMSWorks Software Private Limited Chennai, India
Software Engineer May 2014 – May 2017

- Developed network discovery and provisioning modules, reducing service activation time.
- Built a Deletion Tracking Tool for telecom objects, improving error traceability and debugging.
- Optimized performance by analyzing memory and thread dumps, improving system efficiency from 80% to 95%.

PROJECTS:

ONF TAPI-Based Transport Network Discovery & Orchestration

- Designed and implemented Spring Boot microservices for ONF TAPI-based transport network inventory management.
- Optimized network discovery with multithreading, reducing discovery time by 50%.
- Integrated Kafka-based event-driven architecture, ensuring seamless communication between microservices.

Provisioning – ResourceAllocation framework

- Built an automated resource allocation framework for optical networks, optimizing bandwidth and reducing network planning efforts.
- Developed fragmentation-aware resource allocation, ensuring efficient utilization of network resources.
- Implemented Kafka-based caching to maintain state persistence across application restarts, improving system resilience.

Inventory Consolidation & Dashboard Metrics

- Designed a statistical inventory module, consolidating periodic inventory data for real-time network analytics.
- Developed metrics, trends, and utilization dashboards, reducing network maintenance efforts by 40%.

Enhanced Telecom Deletion framework

- Developed a deletion tracking system, enabling auditability of deletion processes for customers.
- Improved telecom object deletion performance using parallel execution and dependency-aware task ordering.

Probe – Application Error Profiler (Error Logging)

- Built an error logging system for reconciliation frameworks using Apache Kafka.
- Enabled proactive issue detection, reducing support team workload by 50% through automation.

Deployment & DevOps

- Containerized microservices using Docker for efficient application deployment.
- Orchestrated deployments using Kubernetes, ensuring high availability, load balancing, and self-healing.
- Automated CI/CD pipelines using Jenkins & GitLab, accelerating deployment cycles.

KEY STRENGTHS:

- *Microservices & API Development:* Expertise in Spring Boot & Spring Cloud for building scalable applications.
- *Team Leadership:* Hands-on experience managing a 4-member development team, including mentoring, reviewing and project coordination.
- *ONF TAPI-Based Implementation:* Strong understanding of NMS/EMS based enterprise application as well as Transport API for network orchestration**.**
- *Event-Driven Systems:* Designed Kafka-based messaging frameworks for real-time data exchange.
- *Cloud-Native Deployment:* Hands-on experience with Docker & Kubernetes for microservices deployment.
- *Database Optimization:* Improved MySQL query performance and reduced DB load by 10-20%.
- *Multithreading & Performance Tuning:* Optimized systems, improving throughput and reducing latency.